Determine by inspection whether the vectors in Exercises 15–20 are linearly independent. Justify each answer.

15. $\begin{bmatrix}5\\1\end{bmatrix}, \begin{bmatrix}2\\8\end{bmatrix}, \begin{bmatrix}1\\3\end{bmatrix}, \begin{bmatrix}-1\\7\end{bmatrix}$

$\boxed{\text{Solution:}}$ No. of vectors $=4>$ No. of entries in each vector $=2$.
By Theorem 8, the vectors are linearly dependent.

17. $\begin{bmatrix}3\\5\\-1\end{bmatrix}, \begin{bmatrix}0\\0\\0\end{bmatrix}, \begin{bmatrix}-6\\5\\4\end{bmatrix}$

$\boxed{\text{Solution:}}$ There is the presence of a zero vector. By Theorem 9, the vectors are linearly dependent.

19. $\underset{u}{\begin{bmatrix}-8\\12\\-4\end{bmatrix}}, \underset{v}{\begin{bmatrix}2\\-3\\-1\end{bmatrix}}$.

$\boxed{\text{Solution:}}$ Let, u be a multiple of v. $\therefore$ There exists $c\in\mathbb{R}$ s.t. $u=cv$.

$$\begin{bmatrix}-8\\12\\-4\end{bmatrix}=c\begin{bmatrix}2\\-3\\-1\end{bmatrix} \Rightarrow \begin{bmatrix}-8\\12\\-4\end{bmatrix}=\begin{bmatrix}2c\\-3c\\-c\end{bmatrix}$$ (Scalar multiplication)

$\therefore 2c=-8 \Rightarrow c=-4$
$-3c=12 \Rightarrow c=-4$
$-c=-4 \Rightarrow c=4$. This is a contradiction. $\therefore$ u is not a multiple of v.

We can similarly show that v is not a multiple of u.
$\therefore$ u and v are linearly independent.

In Exercises 21 and 22, mark each statement True or False. Justify each answer on the basis of a careful reading of the text.

21) a. The columns of a matrix A are linearly independent if the equation $Ax=0$ has the trivial solution.

$\boxed{\text{Solution:}}$ On Page 43, under the section "Homogeneous Linear Systems" it's mentioned that $Ax=0$ always has atleast the trivial solution (0 vector) as one of the solutions.